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MERN.

1.) Attribute:

An HTML attribute provides addition information about an HTML element. It control the behaviour.

Types:

1.) Global Attributes,
2.) Element specific Attributes.
3.) Boolean Attributes.

2.) CSS layout.

It is used to arrange the positions of HTML elements.

Types:

1.) flex
2.) Grid
3.) position.

3.) Programming language.

It is used to communicate to a computer and perform tasks. It helps to solve problem and logical thinking.

Ex:
JS, C, C++, Python.

4.) Javascript.

It is high-level programming lang. It is mainly used to develop web applications.

features.

1.) dynamic.
2.) object based
3.) Asynchronous
4.) DOM

5.) JS functions in JS.

It is a block of code and also to be reused for the specific task. It executes when it is called.

Types:

1.) Function declaration (or) named fun.
2.) Asynchronous function.
3.) Arrow Fun. 4.) call back.

6.) Fetch API in JS.

It is the one of the modern JS feature, used to send the HTTP request and receive the data from server or APIs.

We use GET, POST, PUT, DELETE to fetch the datas.

7.) React JS.

It is a open-source JS library, developed by facebook (now called meta).

features:

-> component based architecture.
-> virtual DOM. -> JSX
-> one way dataflow.

8.) ES6.

1.) let and const
-> Arrow function
-> Destructuring.
-> spread operator.
-> rest parameter
-> classes. -> promises

ES6+:

-> Async function
-> rest & spread for objects.
-> optional chaining & nullish coalesing
-> class fields.

9.) npm

	npx.
used to install and manage packages	used to execute package.
Installs dependencies	Runs packages commands
commonly used with npm. install	commonly used with npx.
can install local of global package	can run locally installed or remotely fetched packages.

10.) 1.) Babel:

It is a JS compiler that converts modern JS and JSX into code that can be understood by supported JS environment.

2.) Esbuild.

It is a fast javascript bundler and build tool written in Go.

3.) webpack:

webpack is a module bundler used to combine JS, CSS, and others.

4.) bundler:

A bundler is a tool that combins multiple files and modules into one or more optimized output files for production.

11.) what is vite:

It is modern frontend build tool used to create and develop web application.

features:

1.) fast development server
2.) Hot module Replacement
3.) ES module.
4.) Framework support

12.) 1.) SWC: -> oxidation compiler.

It is a collection of high performance TS-based typescript & tools written in Rust.

2.) Rollup:

Rollup is a JS module bundler used to compile multiple modules into optimized output files.

3.) Rolldown:

It is a JS bundler written in Rust. It is designed to provide high performance and compatibility with Rollup's ecosystem.

13.) npm run dev: -> used to start the development server

npm start: used to execute the start script defined in package.json.

npm run build: used to create an optimized production version of the application.

14.) TSX:

TSX is stands for Javascript XML. It is a syntax extension used in React that allows developers to write HTML like code inside Javascript.

TS	TSX
syntax extension JS	programming language.
uses HTML-like syntax	uses TS syntax
mainly used with React	used for front-end back-end development.
requirement in typical TSX	executes directly.
makes UI code	handles logic.